

## Data Collection and Preprocessing Phase

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 01 August 2026                             |
| <b>Team ID</b>       | _____                                      |
| <b>Project Title</b> | AI Adoption and Impact in Global Education |
| <b>Maximum Marks</b> | 3 Marks                                    |

### Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

| Data Source | Data Quality Issue                                                                                                                                          | Severity | Resolution Plan                                                                                                                                                                                                              |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset     | Missing values may be present in fields such as AI usage, adoption rates, internet penetration, education index, or policy indicators.                      | Moderate | Profile null values using pandas/Tableau. For numerical fields, use an appropriate imputation method only when justified; otherwise retain nulls or exclude affected records from calculations. Document the treatment used. |
| Dataset     | Duplicate records may cause the same country, demographic group, or observation to be counted more than once.                                               | Moderate | Identify exact duplicates using duplicate-row checks and remove confirmed duplicate records. Validate key fields before removing non-identical records.                                                                      |
| Dataset     | Inconsistent categorical values may occur in country, region, demographic, or AI tool names because of spelling, capitalization, or formatting differences. | Moderate | Standardize categorical values using mapping/dictionary rules. Trim whitespace, normalize capitalization, and map equivalent labels to a single standardized value.                                                          |
| Dataset     | Numeric fields may contain incorrect formats or non-numeric characters, which can prevent accurate calculations and visualizations.                         | High     | Convert relevant fields to numeric data types after removing valid formatting characters. Flag values that cannot be converted and investigate them before analysis.                                                         |
| Dataset     | Potential outliers or unrealistic values may occur in adoption percentages, daily usage time, internet penetration, or index values.                        | High     | Apply range checks and descriptive statistics to identify suspicious values. Verify against the source where possible and correct confirmed errors; do not remove legitimate extreme observations without justification.     |

|         |                                                                                                                              |     |                                                                                                                                                                       |
|---------|------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset | Country and region information may not follow a consistent geographic classification, which can affect regional comparisons. | Low | Create a standardized country-to-region mapping and validate that each country belongs to the intended region before building geographic and regional visualizations. |
|---------|------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

*Note: The report identifies the main data-quality checks and resolution plans for the project dataset. Exact counts of missing values, duplicates, and outliers should be updated after the final dataset is inspected.*